

Diamond Sales Analysis

Data Analyst Final Project — University Capstone

Dataset

4,498 Records · 19 Variables

Period

2022 – 2024

Markets

6 Global Locations

Scope

EDA · Hypothesis Testing · ML · Clustering

Instructor: Ofer David

Student: Shai Levi Altshtein **ID: 324290428**

Made with GAMMA

Dataset Overview & Project Objectives

$3,959

Average Price

Median: $2,430

41.3%

Ideal Cut Share

Dominant cut grade in inventory

60%

Physical Channels

Showroom + Boutique

Sales Channel Distribution

- Showroom 34.2% — largest physical channel
- Boutique 25.6% — in-store channel
- Online 25.0% — digital sales channel
- Wholesale 15.2% — bulk and trade channel

Business Context & Dataset

Diamond sales transactions spanning **6 global luxury markets** — New York, London, Tokyo, Dubai, Paris, and Mumbai. The raw dataset comprised 4,522 records across 19 columns, reduced to **4,499 clean records** after preprocessing.
Key Variables: The Four Cs (Carat, Cut, Color, Clarity) · PriceUSD · AppraisalUSD · DiscountPercent · SalesChannel · Certification · CustomerSatisfaction · DaysOnMarket

Project Objectives

01

Explore pricing drivers via Exploratory Data Analysis

02

Test statistical hypotheses on quality grade effects

03

Build a predictive ML model for diamond price estimation

04

Segment the market using K-Means clustering

Data Preprocessing: From Raw to Reliable

A rigorous preprocessing pipeline ensures downstream analysis and modeling use clean, consistent data. The result is fewer errors, less bias, and more trustworthy conclusions.

Preprocessing Summary

Action Taken	Records Affected	Issue
Converted to numeric	4 columns (Carat, PriceUSD, AppraisalUSD, DaysOnMarket)	Incorrect data types
Standardized (strip + title case)	Cut, SalesChannel, StoreLocation, Certification	Category inconsistencies
Removed	2 records	Invalid Carat (=99)
Set to NaN → imputed	17 records	Invalid values (Carat ≤0, Height ≤0, Satisfaction >10)
Median / 'Unknown' imputation	DepthPct 4.5%, CustomerSatisfaction 4.0%, Certification 3.5%	Missing values
Removed	22 fully identical rows	Duplicate rows
Ready for EDA	**4,498 rows · 19 columns · 0 missing values**	**Final dataset**

Key Cleaning Operations

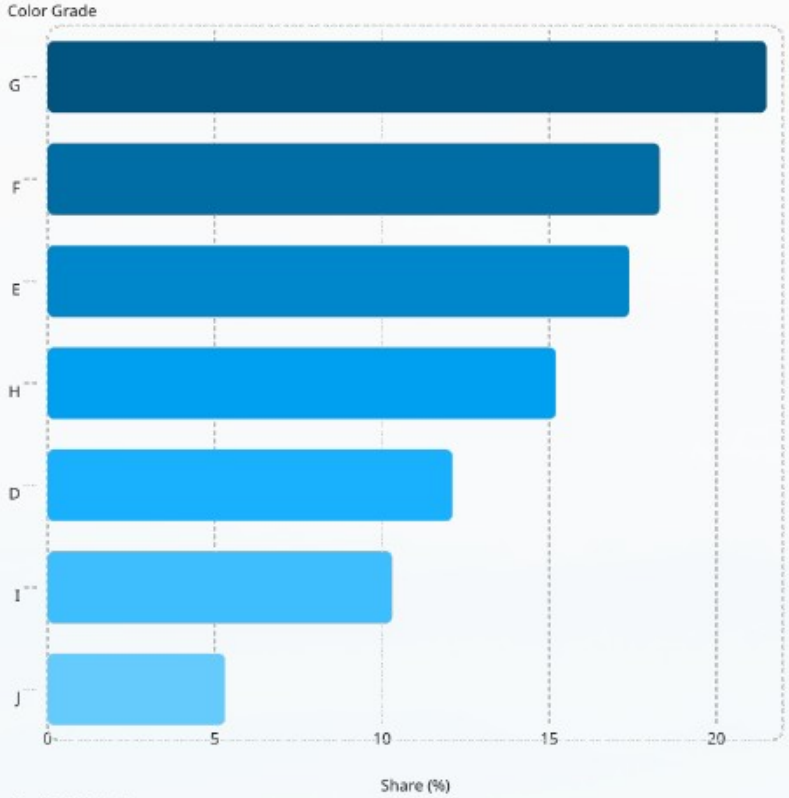
- Format errors fixed: text-based Carat and PriceUSD converted to numeric
- Category standardization: Cut, SalesChannel, StoreLocation, and Certification unified
- Date processing: SaleDate converted to datetime format (dayfirst=True for DD/MM/YYYY values) and validated for time-based analysis
- Outlier removal: 2 invalid Carat = 99.0 records removed
- Missing values: numeric median imputation; categorical values set to Unknown
- Duplicates removed: 22 identical rows, including DIA62001 (same sale date written in two formats)

Exploratory Data Analysis: Key Statistics

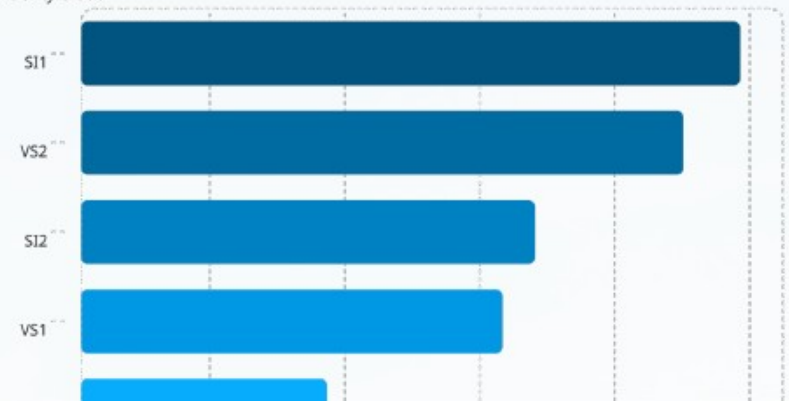
Inventory skews toward mid-range quality

SI1/VS2 clarity and G/F color dominate, making the assortment well-suited for mainstream premium retail positioning.

Color Distribution



Clarity Distribution



Price & Carat Distribution

Both distributions are **right-skewed**, with a small number of large, high-value diamonds driving the upper tail. Mean price ($3,959) exceeds median ($2,433), confirming positive skew.

0.80ct

Avg. Carat

Median: 0.70 ct — most stones are small-to-medium

8.1

Median Satisfaction

Out of 10 — consistently high across all channels

Geographic Distribution

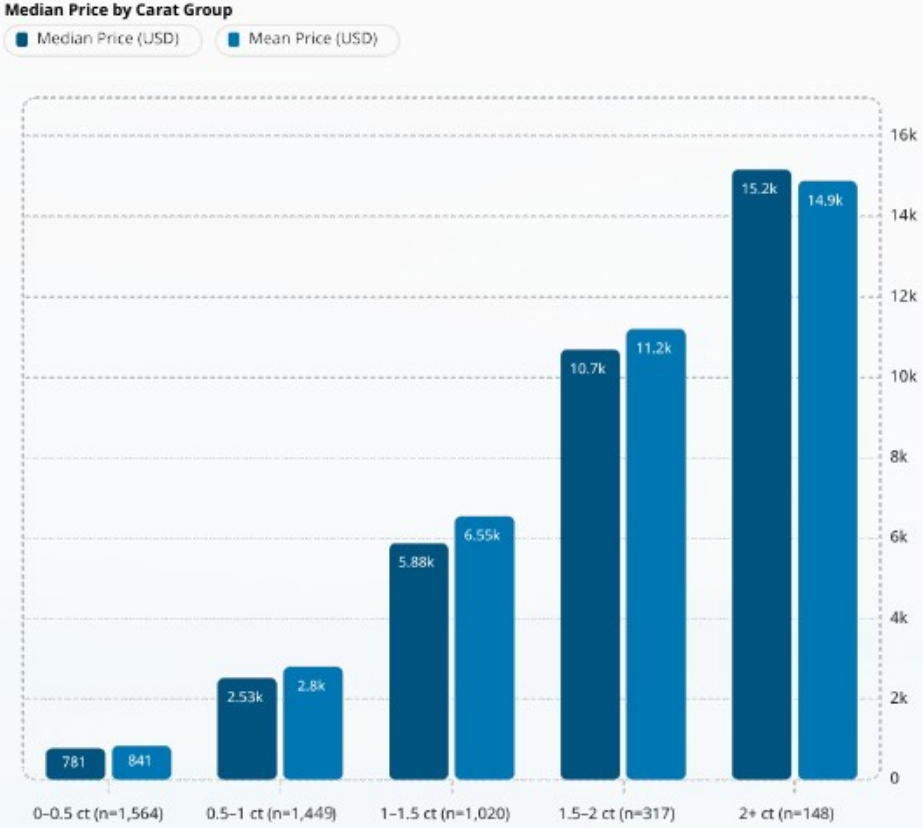
- **New York** — 988 sales (largest market)
- **London** — 897 sales
- **Tokyo** — 751 · **Dubai** — 714 · **Paris** — 618 · **Mumbai** — 440

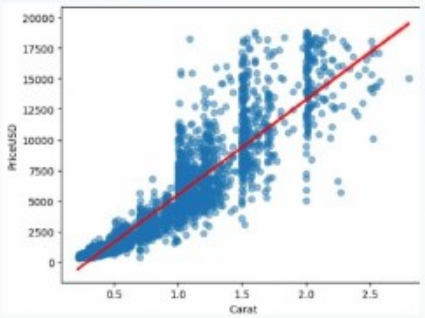
Certification Coverage

- GIA **40.9%**
- IGI **27.0%**
- Uncertified **14.6%**
- HRD **13.9%**
- Unknown **3.6%**

Carat Weight is the Strongest Price Driver

Median Price by Carat Group





Correlation Analysis

0.919

Pearson r

Very strong positive linear relationship between carat and price

0.96

Spearman ρ

Confirmed by rank-based non-parametric analysis

Key Observations

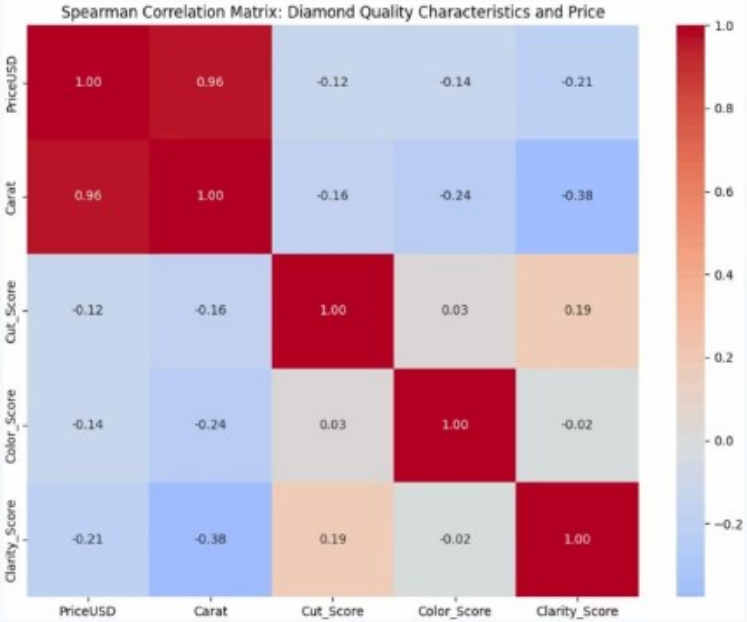
- Price variance increases with carat weight.
- Quality matters more for larger diamonds.
- Large stones drive the upper price tail.

Carat is the primary pricing lever. Quality characteristics become increasingly critical for accurate large-stone valuation.

Diamond Quality: Cut, Color & Clarity Impact on Price

Counterintuitive finding: Lower-quality grades show higher raw median prices. This pattern is largely explained by carat confounding and should not be interpreted as an independent quality effect.

Quality Grade Visualization



Ideal cut is the most common cut category (41.3%), while VVS1 and IF are relatively rare. Pricing analyses must account for carat weight before interpreting quality-grade effects.

Root Cause: Carat Confounding

Lower-quality diamonds are systematically **heavier** — carat dominates the raw price signal and masks quality grade effects:

Higher Grade Avg. Carat	Lower Grade Avg. Carat	Quality Dimension
Ideal: **0.70 ct**	Fair: **1.14 ct**	Cut
D: **0.66 ct**	J: **1.17 ct**	Color
IF: **0.52 ct**	I1: **1.26 ct**	Clarity

Spearman Correlations with Price

Carat ρ = 0.96
Dominant predictor

Clarity ρ = −0.21
Weak negative

Color ρ = −0.14
Weak negative

Cut ρ = −0.12
Weak negative

Sales & Market Analysis

Geographic Markets

New York
988 sales · Largest market

London
897 sales

Tokyo
751 sales

Dubai
714 sales

Paris
618 sales

Mumbai
440 sales

Operational Metrics

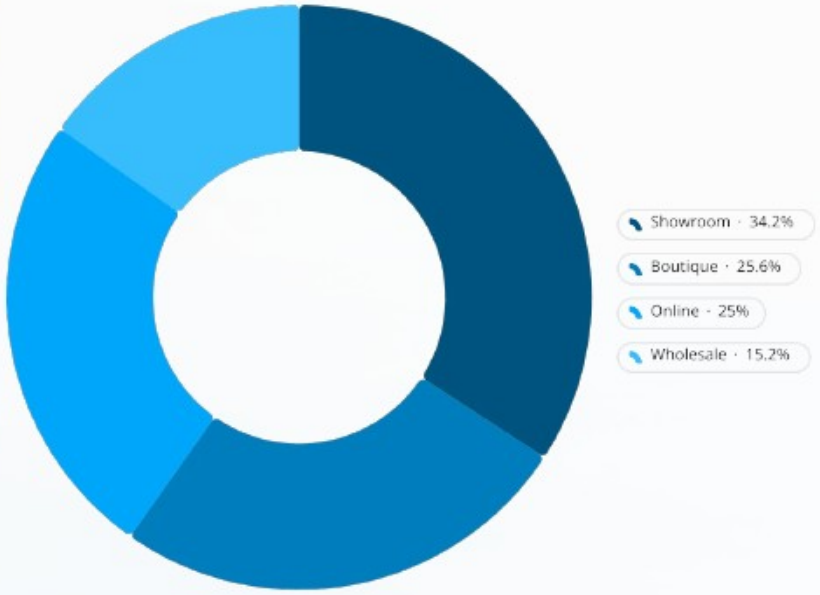
Discount Range
−16.4% to +22.9% — wide spread signals varied pricing strategies across channels and segments

Days on Market
Median ~28 days — relatively fast turnover, indicating healthy demand across the portfolio

Customer Satisfaction
Median 8.1/10 — high and consistent; most common scores: 8.1, 8.2, 7.7, 8.0

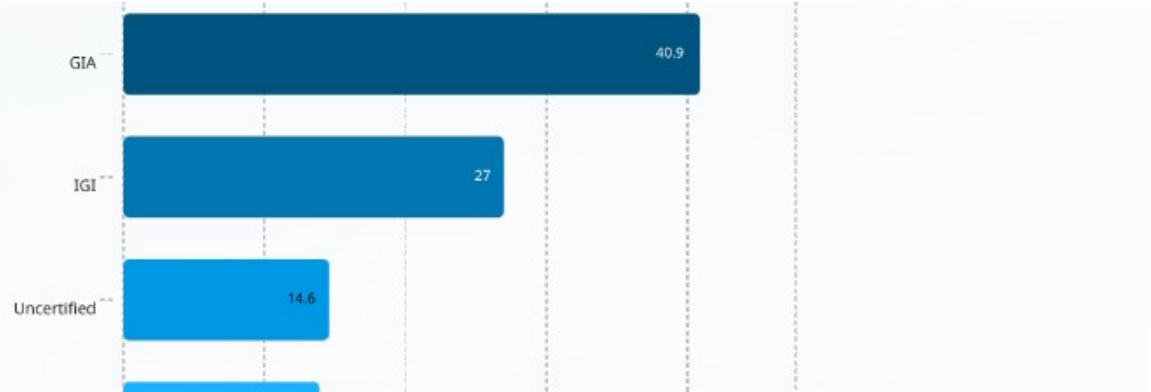
Satisfaction is high overall; causal marketing effects were not tested.

Sales Channel Performance



Showroom and Boutique together account for **~60% of sales**, showing that physical experience still drives high-value purchases. Online at 25% remains the clearest digital growth opportunity.

Physical channels represent 59.8% of sales; Online represents 25.0%. The dataset does not establish channel growth over time.



Statistical Hypothesis Testing

DECISION: Fail to Reject H_o

Welch's t-test · Premium vs. Standard diamonds · 1–1.5 carats

Research Question

Is the difference in mean selling price between Premium and Standard diamonds within the 1–1.5 carat range statistically significant?

H_o: No statistically significant mean-price difference.
H_1: A statistically significant mean-price difference exists.

Test Design

Levene's test: p = 0.358
Welch's t-test: t = 1.908 · p = 0.0568
Significance level: α = 0.05
Cohen's d = 0.126 (very small effect)

Observed Results

Premium	Standard	Difference	
n = 675 · Mean = $6,652.72	n = 345 · Mean = $6,341.86	$310.87	Not significant at α = 0.05

Conclusion: Premium diamonds were slightly more expensive within this range, but the difference was not statistically significant and the effect size was very small. Pricing should consider multiple characteristics, not only this classification.

Machine Learning: Linear Regression Price Prediction

Final Selected Model: Log-Transformed Linear Regression

80/20 split · random_state = 42 · 4,498 cleaned records · 5-fold cross-validation

Baseline Linear Regression

R^2 0.9253
MAE $743.78
RMSE $1,106.92
MAPE 41.98%
Negative predictions 76
Mean CV R^2 0.9183

Final Log-Transformed Model

R^2 0.9605
MAE $434.22
RMSE $804.81
MAPE 10.70%
Negative predictions 0
Mean CV R^2 0.9505 · SD 0.0040

Model Inputs & Encoding

Carat · Cut · Color · Clarity · DepthPct · TablePct
LengthMm · WidthMm · HeightMm · Certification
One-Hot Encoding produced 27 model features.

Interpretation & Future Work

Permutation importance ranked Carat first, followed by LengthMm. Compare against Regularized Regression, Random Forest and Gradient Boosting; validate with external market data.

The final model improved every metric, reduced MAPE from 41.98% to 10.70%, and eliminated negative predictions. Use it to support—not replace—professional appraisal.

Diamond Market Segmentation: K-Means Clustering

K-Means Market Segmentation — k = 3

StandardScaler · 7 numerical features · 4,498 diamonds · Elbow Method

Entry-Level Diamonds

2,079 · 46.22%

0.40 ct average

$1,056.45 average price

Small carat weights, low selling prices and smaller physical dimensions.

Mid-Value Diamonds

1,800 · 40.02%

0.95 ct average

$4,582.40 average price

Moderate carat weights, intermediate selling prices and medium physical dimensions.

Premium Diamonds

619 · 13.76%

1.69 ct average

$11,895.39 average price

Large carat weights, high selling prices and larger physical dimensions.

Features: Carat · PriceUSD · DepthPct · TablePct · LengthMm · WidthMm · HeightMm

Silhouette 0.381 · Davies–Bouldin 1.042

K-Means identified three broad, interpretable market segments with moderate separation and some overlap. Treat them as decision-support categories—not strict classifications—and add profiling variables before assigning customer behavior or marketing strategies.

Key Findings & Business Recommendations

Core Analytical Findings

1. **Carat is the dominant price driver**

   Pearson r = 0.919; Spearman ρ = 0.96.

2. **Quality comparisons require carat control**

   Raw grade-price patterns are confounded by size.

3. **Physical channels dominate**

   Showroom + Boutique = 59.8%; Online = 25.0%.

4. **Three broad diamond segments identified**

   Moderate separation; clusters are not customer profiles.

Evidence-Based Recommendations

Model	Use the final log model for pricing support; compare it with regularized and tree-based models.
Inventory	Use clusters for product positioning and inventory organization, with caution near cluster boundaries.
Profiling	Add discount, certification, channel and satisfaction before segment-specific campaigns.
Validation	Use external market data and cluster-stability analysis before operational deployment.

KEY TAKEAWAY

Diamond pricing is primarily driven by carat and physical dimensions. Quality effects require controlled analysis; the final log-transformed model provides strong decision support; and the three K-Means clusters offer broad product categories rather than customer personas.

Tableau Appendix: Diamond Color Distribution

965

G-color diamonds

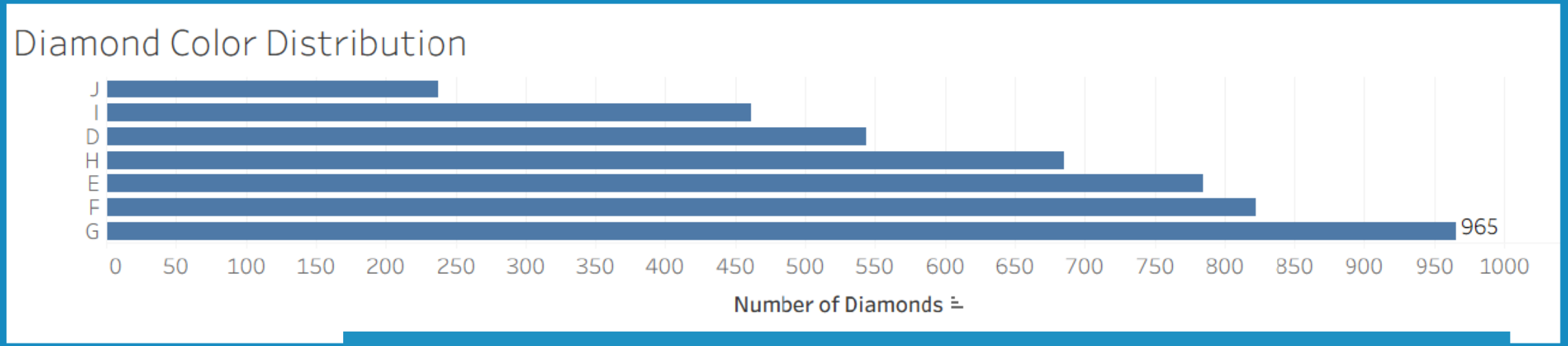
G is the largest color category, representing 21.45% of the cleaned dataset.

69.2%

D–G share

Colors D through G together account for 3,114 of the 4,498 cleaned records.

The distribution is consistent with the notebook results (n = 4,499) and supports prioritizing the high-volume G–F–E range.



The Tableau ranking matches the notebook frequency table (n = 4,498) and describes the observed inventory mix.

Tableau Appendix: Diamond Clarity Distribution

1,098

SI1 diamonds

SI1 is the largest clarity category, accounting for 24.41% of the cleaned dataset.

46.7%

SI1 + VS2 share

SI1 and VS2 together represent nearly half of all observations.

The Tableau ranking exactly matches the notebook frequency table and highlights a portfolio centered on mid-market clarity grades.

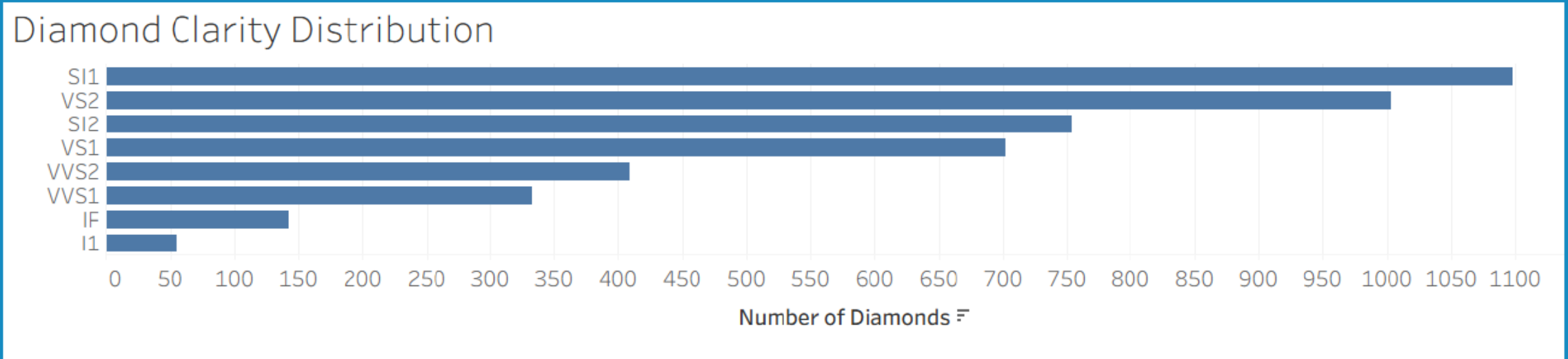


Tableau Appendix: Diamond Price vs. Carat

0.919

Pearson correlation

Price rises strongly with carat weight, confirming the notebook's strongest linear relationship.

0.96

Spearman correlation

The rank-based result confirms a very strong monotonic relationship between size and price.

This view supports the notebook conclusion: carat is the primary price driver, while cut and other quality factors provide secondary differentiation.

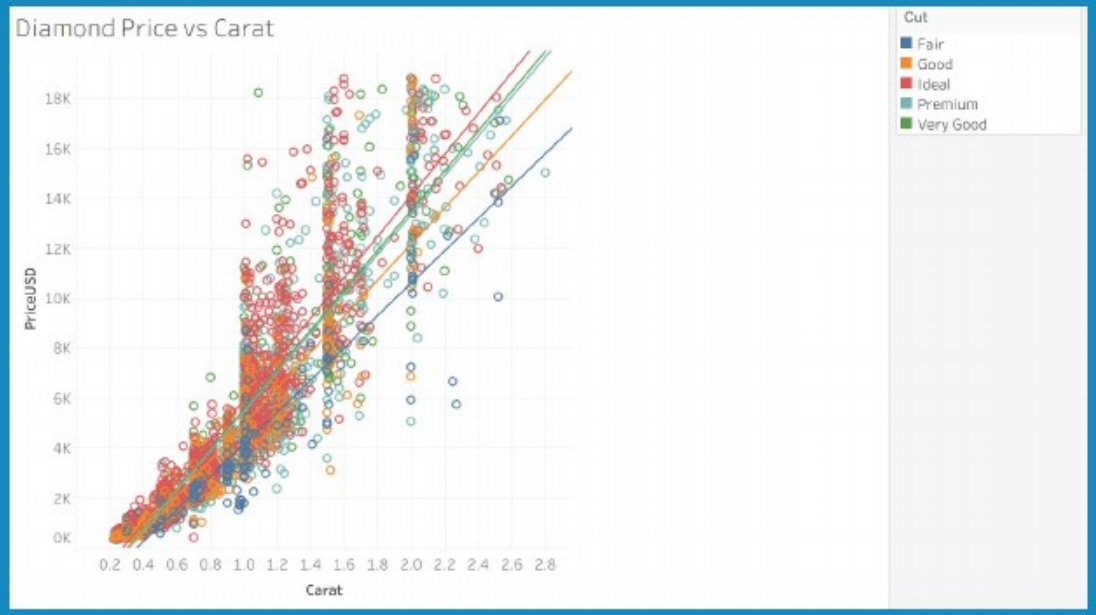


Tableau Appendix: Diamond Certification Distribution

40.9%

GIA-certified

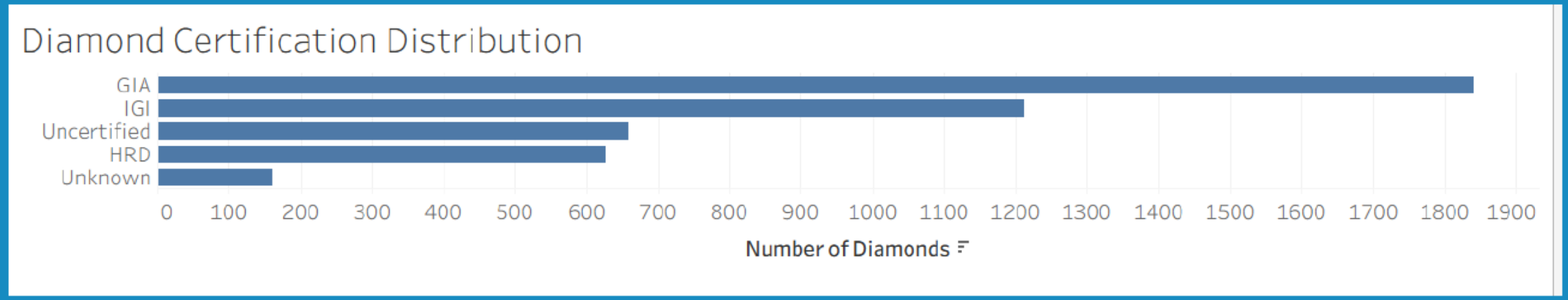
GIA is the leading certification category in the cleaned dataset.

27.0%

IGI-certified

IGI is the second-largest category; together, GIA and IGI cover most certified inventory.

The Tableau chart is consistent with the presentation's certification breakdown and reinforces GIA as the strongest certification signal.



The chart matches the notebook certification breakdown. GIA is the most common category; this alone does not establish a price premium.

Tableau Appendix: Diamond Sales Channel Distribution

1,538

Showroom sales

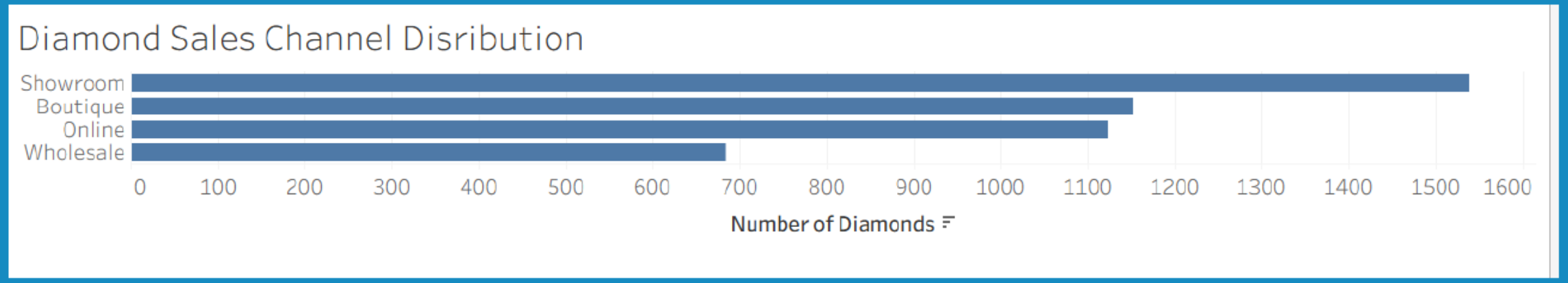
Showroom is the largest channel, representing 34.19% of all cleaned records.

59.8%

Physical premium channels

Showroom and Boutique together account for nearly 60% of sales.

The chart exactly matches the notebook channel summary and supports the deck's finding that physical channels dominate while online remains a growth opportunity.



The chart matches the notebook channel summary: physical channels account for 59.8% of sales, while Online accounts for 25.0%.